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Complete the blanks in the following question with the appropriate answer.

A system has X MB, Y way set associative cache with block size Z Bytes. Processor generates M bit addresses. In addition to tag, there are 2 valid bit, 1 replacement bit and 1 modified bit.

|   |    |
|---|----|
| X | 2  |
| Y | 16 |
| Z | 64 |
| M | 32 |

The size of the tag array is \_\_\_\_\_ KBytes

Submit Answer & Continue

all at one point telegram channel @oahelp

#### 14. DDR DRAM Fetch Cycles

Complete the blanks in the following question with the appropriate answer.

Find the number of cycles to fetch X Bytes of data from a DDR DRAM with bus width equal to Y bits  
Note: Assume no constrain on Burst Length

|   |    |
|---|----|
| X | 8  |
| Y | 16 |

Number of cycle = \_\_\_\_\_

Submit Answer & Continue

DELL



ALL



9

10

11

12

13

14

15

## 12. Vowels never together

Complete the blanks in the following question with the appropriate answer.

Come up with the number of words that can be formed from the letters of word W (listed below) so that the vowels never come together.

|   |       |
|---|-------|
| W | TEGRA |
|---|-------|

Number of words \_\_\_\_\_

Submit Answer & Continue

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|    |                                       |     |
|----|---------------------------------------|-----|
|    | Memory Bytes per mem cycle X          | 128 |
| 13 | Request size A Bytes                  | 64  |
| 14 | Fc - CPU freq                         | 3   |
| 15 | Fm - Memory Freq                      | 2   |
|    | Average time of each request M cycles | 100 |

The Maximum Memory Bandwidth supported is \_\_\_\_\_ GBps

Note : Enter 1 decimal place and round it to nearest value. Ex: If the value is 2.63 Mbytes then type 2.6. If answer is 4, then just type 4 with no decimal.

The throughput that can be achieved as seen from core is \_\_\_\_\_ GBps

Note : Enter 1 decimal place and round it to nearest value. Ex: If the value is 2.63 Mbytes then type 2.6. If answer is 4, then just type 4 with no decimal.

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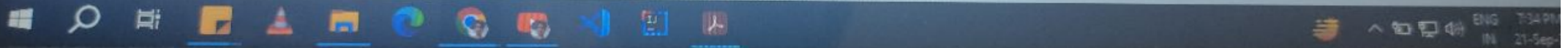
|    |     |
|----|-----|
| K2 | 4   |
| D  | 130 |
| B  | 6   |

For the given values of K1, K2, D, B, the total distance travelled by the ball till Raya and Toya cross each other is \_\_\_\_\_ meters.  
[Note: Please roundoff answer to the nearest integer. For example :

If the answer is 3.5 m, enter 4

if the answer is 3.49 m, enter 3]

Submit Answer & Continue



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```
{
static int count = 0;
count = count + index;
return (count);
}
```

```
main ()
{
int i,j;
int k; // Assume k is initialized to below value
for (i = 0; i <= k; i++)
j = incr(i);
}
```

|   |   |
|---|---|
| k | 7 |
|---|---|

Value of j \_\_\_\_\_

Submit Answer & Continue



DELL





### 13. Recursive Function

Complete the blanks in the following question with the appropriate answer.

$f(x+2) = f(x) + f(x+1)$ , for all positive integers  $x$ .

$f(11) = M$

$f(15) = N$

|   |    |
|---|----|
| M | 10 |
| N | 95 |

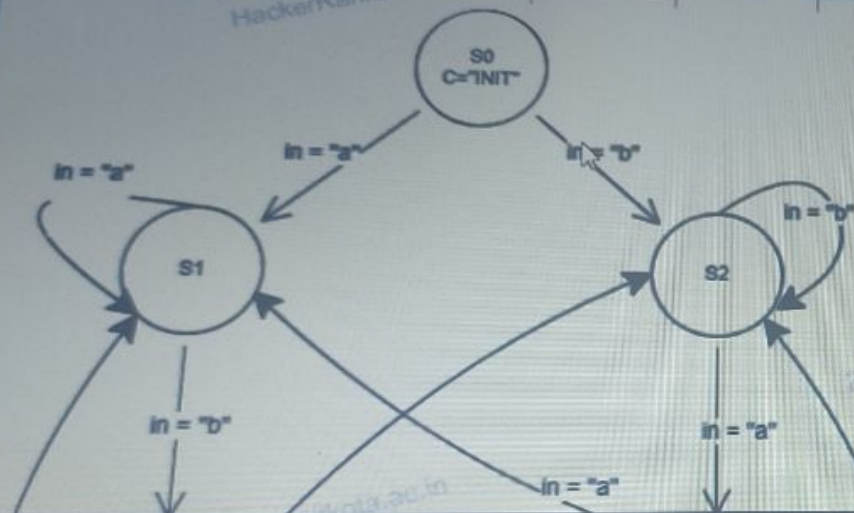
For the given values of  $N$  and  $M$ , value of  $f(10)$  is \_\_\_\_\_

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Complete the blanks in the following question with the appropriate answer.

Given below FSM which works on a single bit input "in", and generates output in the form of an 8 bit state variable "C".  
The value of C is initialized to "INIT" in the idle state S0.  
FnX and FnY are operations performed on the variable C at the FSM transitions S5 -> S6 and S6 -> S2 respectively (as shown in the state diagram)

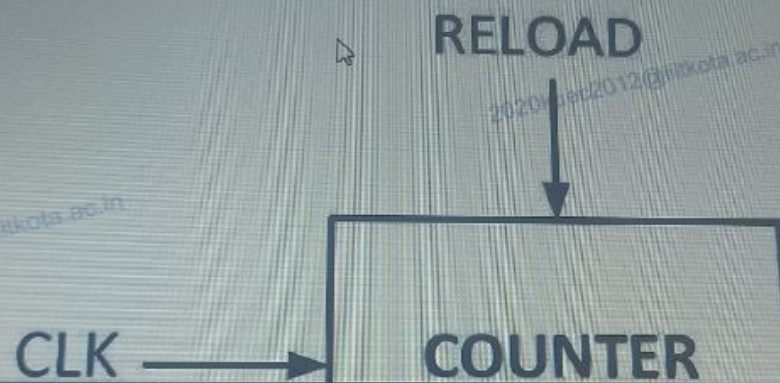


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The counter shown in the diagram below has the following characteristics:

1. Counter increments by value "A" every "B" clock cycles.
2. When "RELOAD" signal in the diagram is asserted, the counter gets initialized to value "0".
3. The RELOAD signal gets asserted every "C" clock cycles and stays asserted for one clock.



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| Parameters |    |
|------------|----|
| A          | 6  |
| B          | 1  |
| C          | 9  |
| X          | 25 |
| Y          | 50 |

What is the counter value sampled at clock-cycle # X: \_\_\_\_\_

What is the counter value sampled at clock-cycle # Y: \_\_\_\_\_

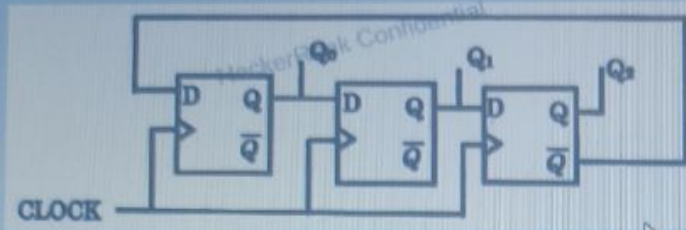
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## 1. Flop output at different clock pulse

Complete the blanks in the following question with the appropriate answer.

Assume that the flops are reset initially, and value of  $(Q_0, Q_1, Q_2)$  is  $(0,0,0)$  at the 0th clock pulse

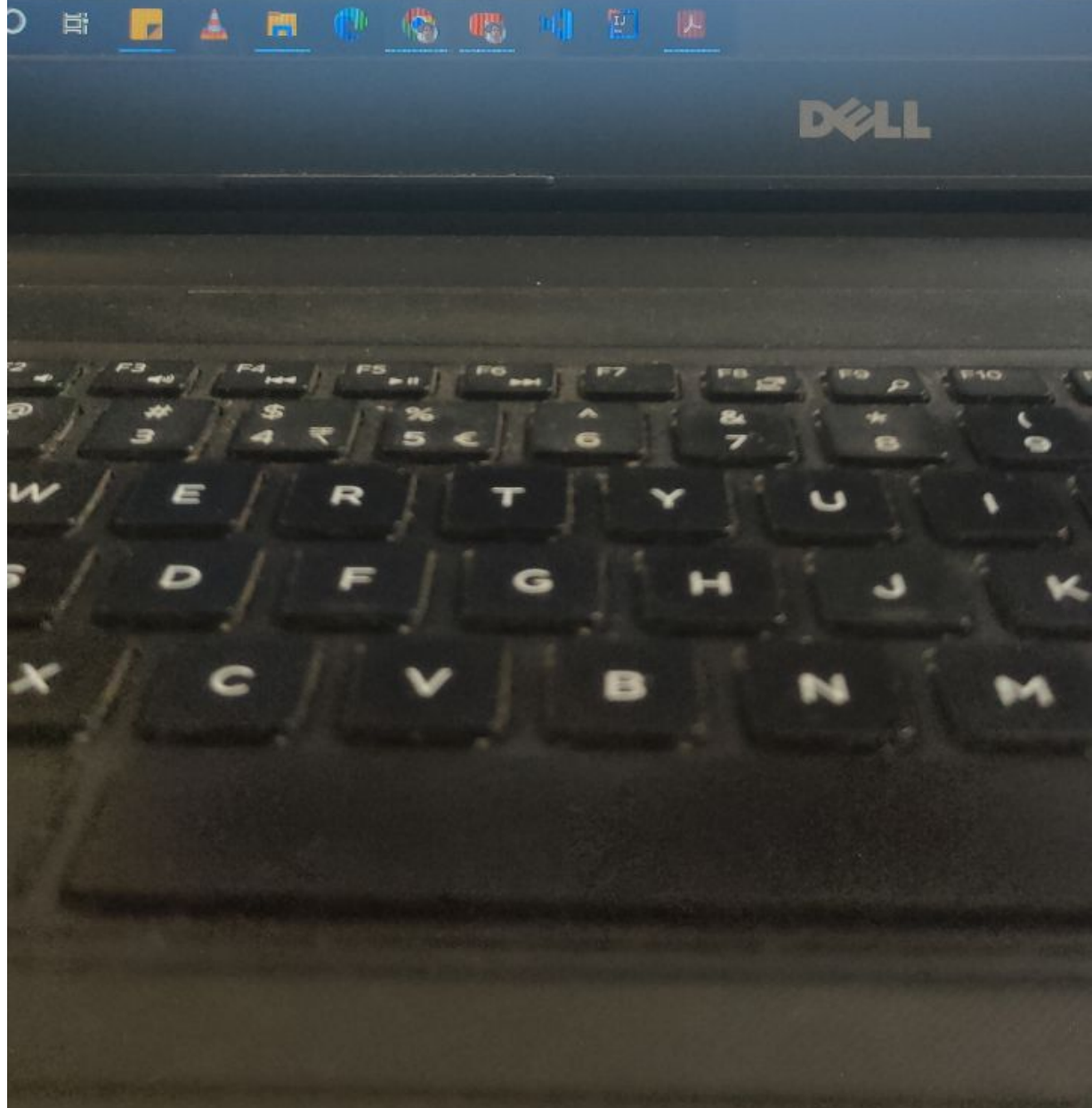


|   |    |
|---|----|
| M | 18 |
| N | 35 |

What is the value of  $(Q_0, Q_1, Q_2)$  at

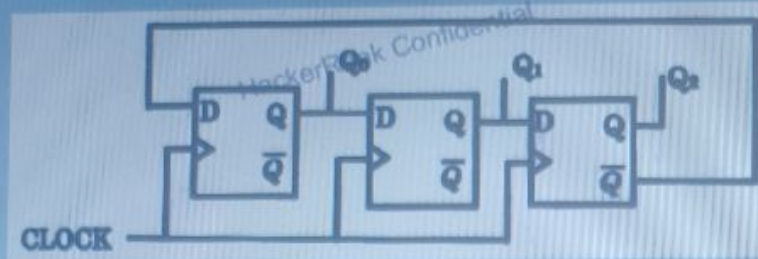
(a) Mth clock pulse : \_\_\_\_\_

(b) Nth clock pulse : \_\_\_\_\_





Assume that the flops are reset initially, and value of (Q0, Q1, Q2) is (0,0,0) at t=0.



|   |    |
|---|----|
| M | 18 |
| N | 35 |

What is the value of (Q0, Q1, Q2) at

(a) Mth clock pulse : \_\_\_\_\_ ;

(b) Nth clock pulse : \_\_\_\_\_ ;

Note: Enter the 3 bits without any coma, for example if the answer is 1,1,0; enter 110

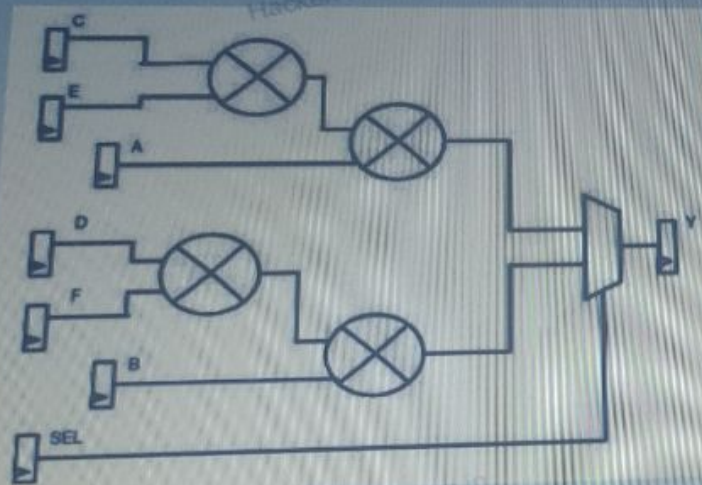
Submit Answer & Continue



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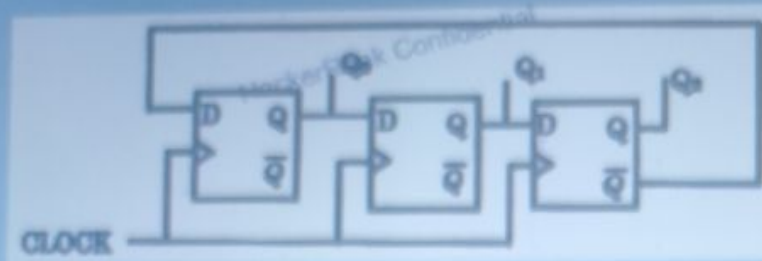
Complete the blanks in the following question with the appropriate answer.

The following block diagram shows an implementation of a computational logic. The area occupied by different components are shown below. Redesign the block diagram to achieve the same functionality with the minimum possible area. What is the total area of your design after minimization?





Assume that the flops are reset initially, and value of  $(Q_0, Q_1, Q_2)$  is  $(0,0,0)$  at  $t=0$



|   |    |
|---|----|
| M | 18 |
| N | 35 |

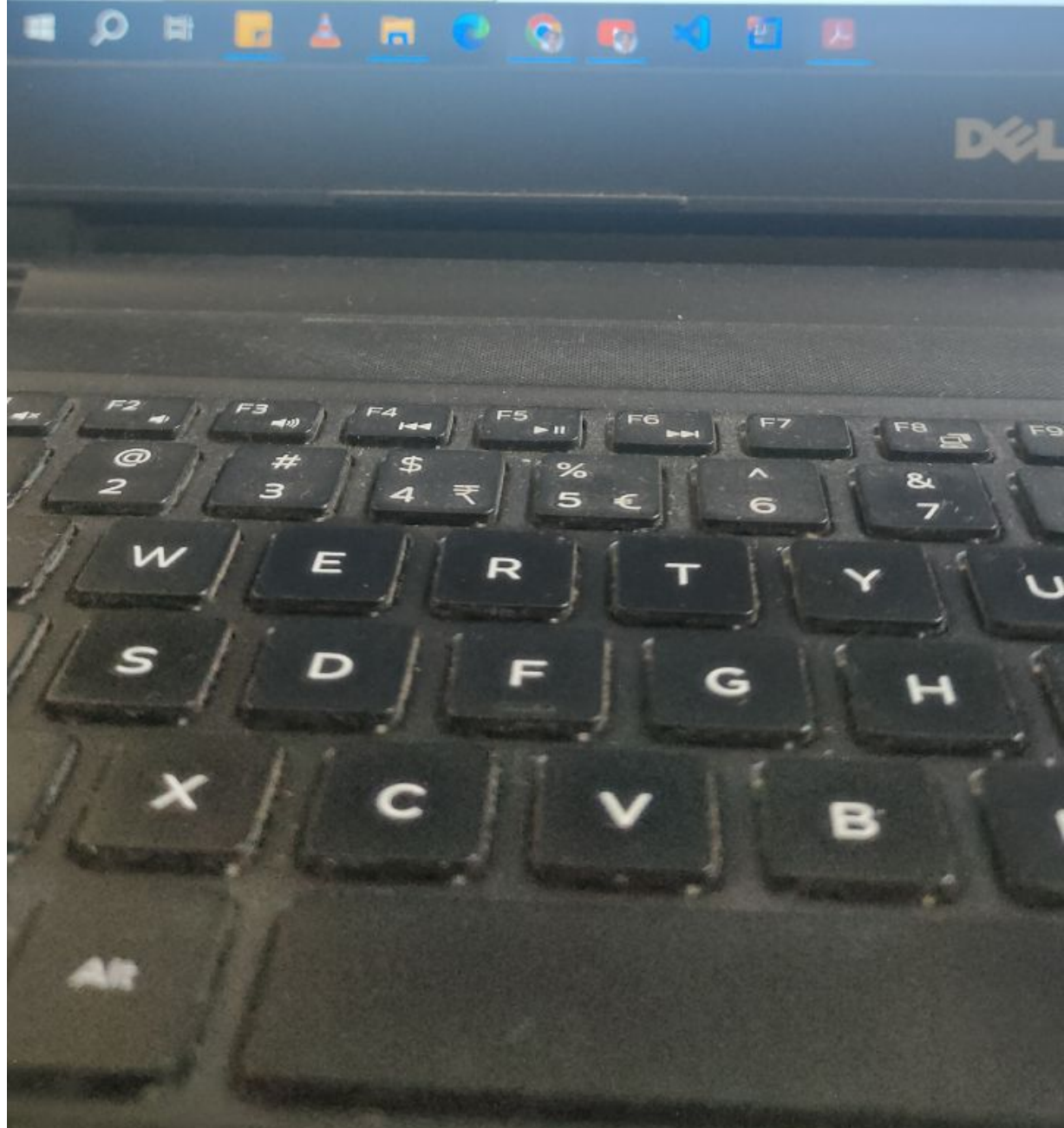
What is the value of  $(Q_0, Q_1, Q_2)$  at

(a) Mth clock pulse : \_\_\_\_\_ ;

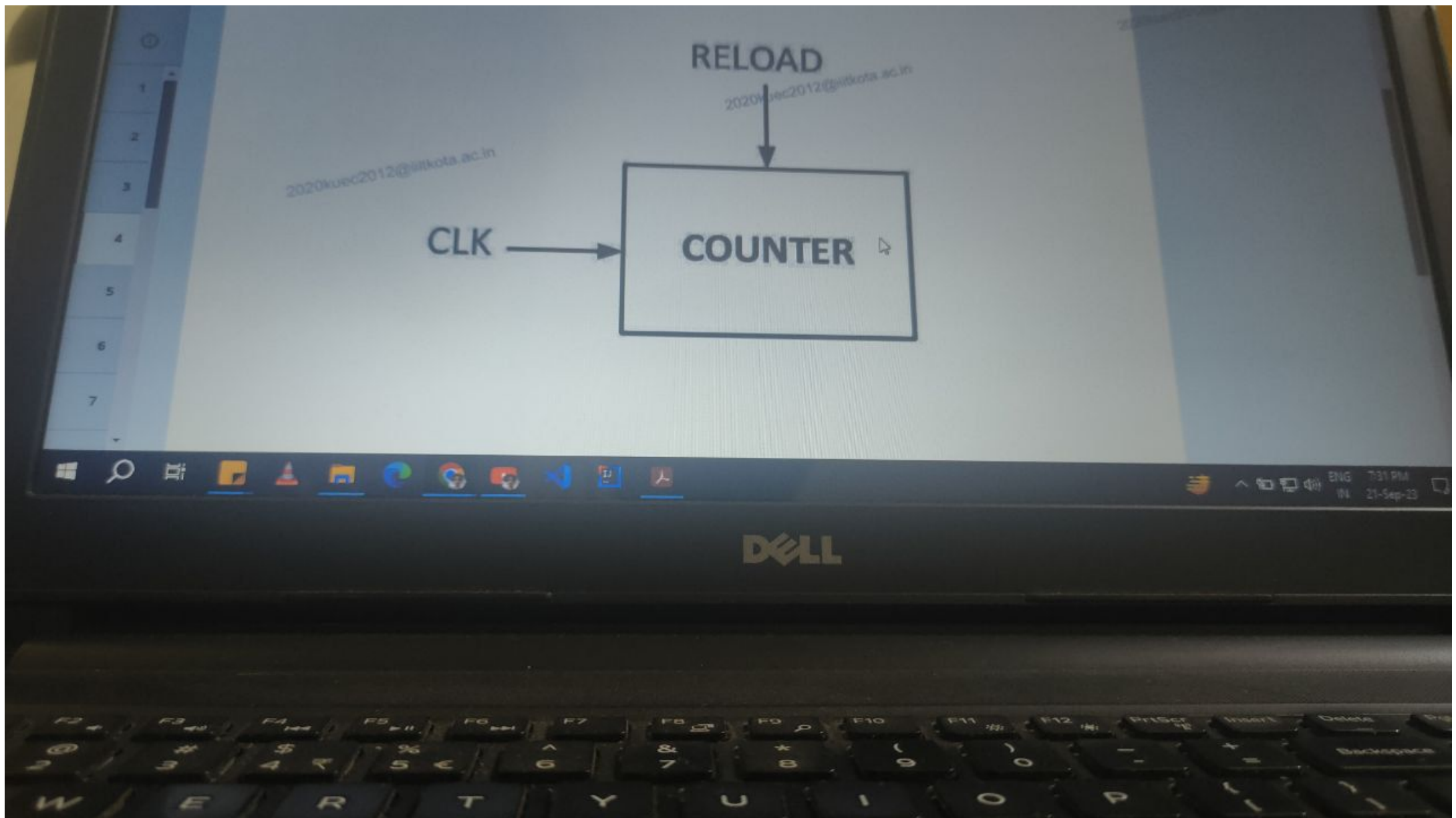
(b) Nth clock pulse : \_\_\_\_\_ ;

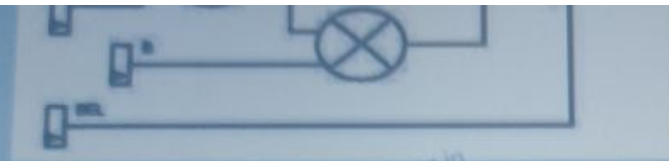
Note: Enter the 3 bits without any coma, for example if the answer is 1,1,0; enter 110

Submit Answer & Continue









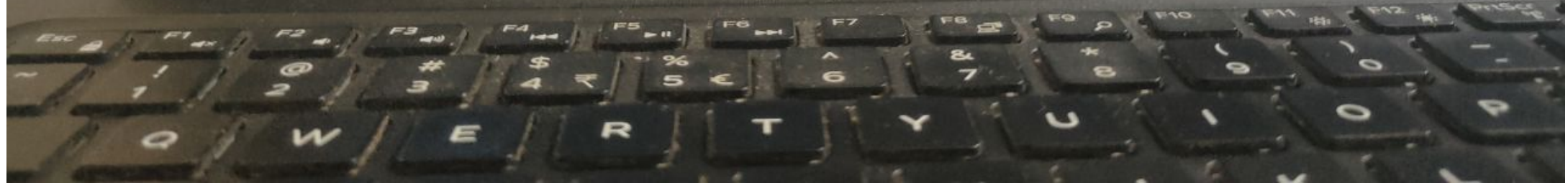
|                               |      |
|-------------------------------|------|
| Multiplier Area (sq. microns) | 3000 |
| Mux Area (sq. microns)        | 150  |
| Register Area (sq. microns)   | 150  |

Total area of the design after minimization (sq. microns) : \_\_\_\_\_

Submit Answer & Continue



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in = "a"

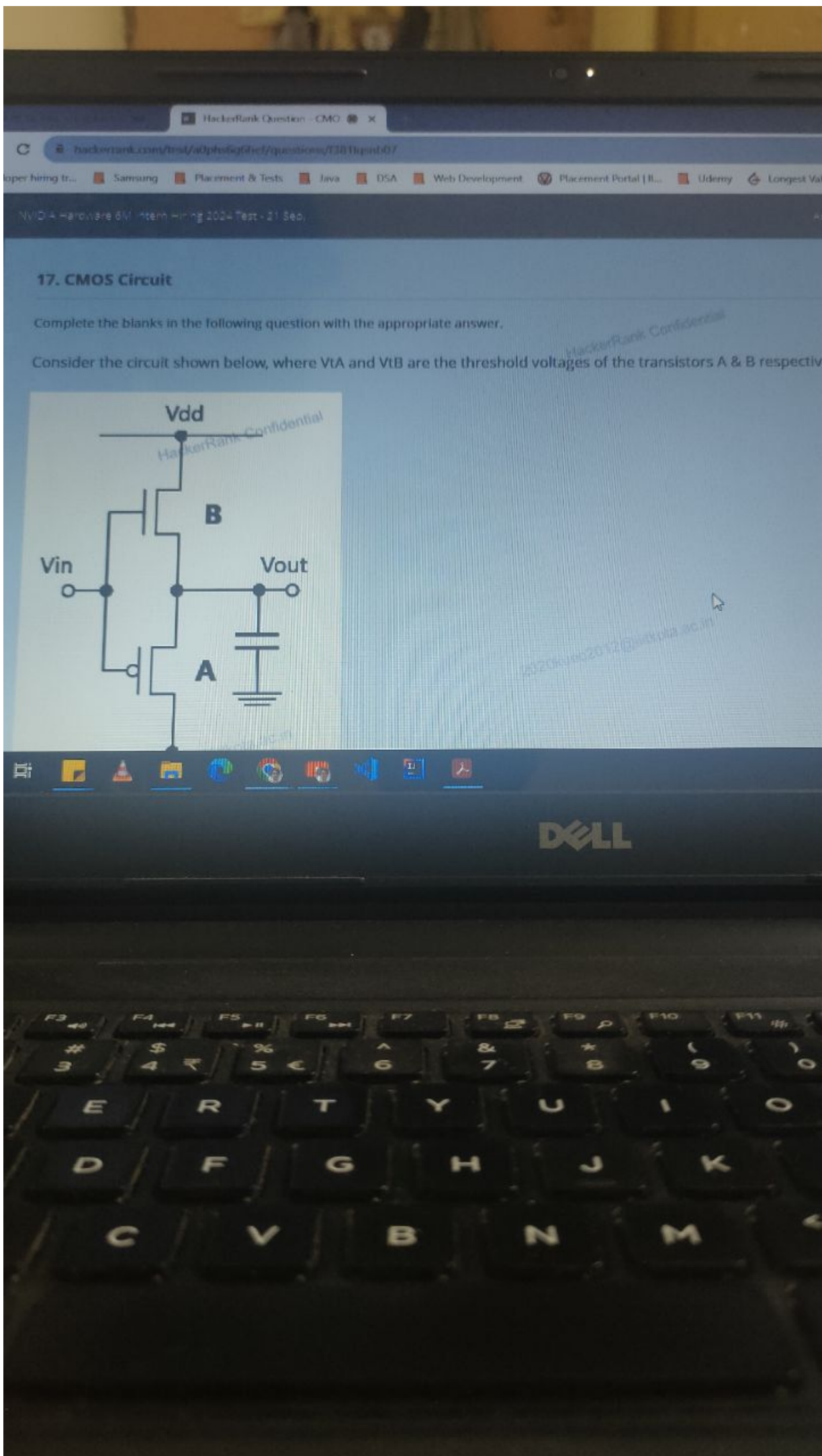
|      |                |
|------|----------------|
| a    | 1              |
| b    | 0              |
| FnX  | C++            |
| FnY  | C--            |
| INIT | 8'd1           |
| SEQ  | 10101001010100 |

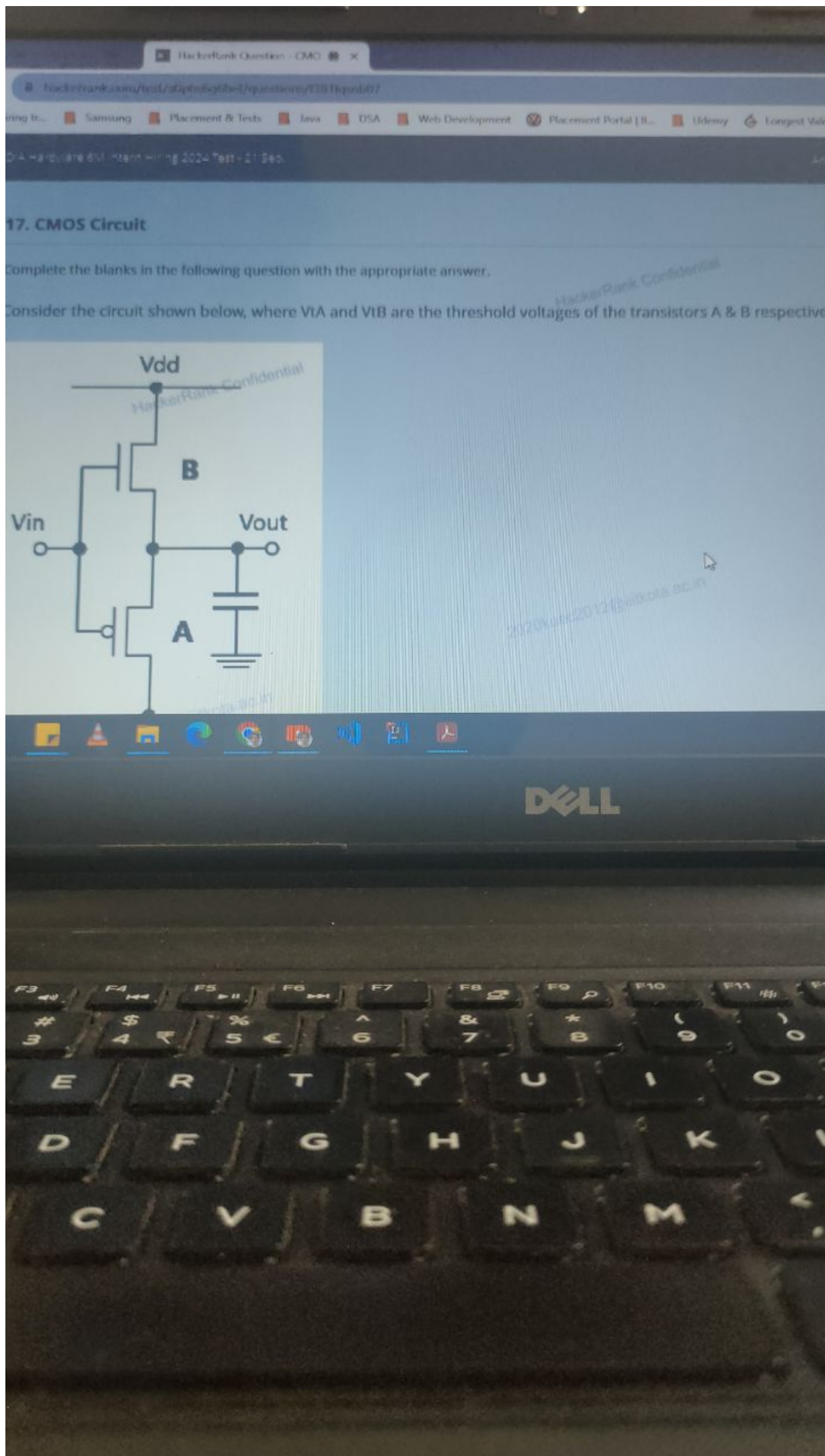
For given parameters and SEQ, starting with leftmost bit identify the value of output C = \_\_\_\_\_

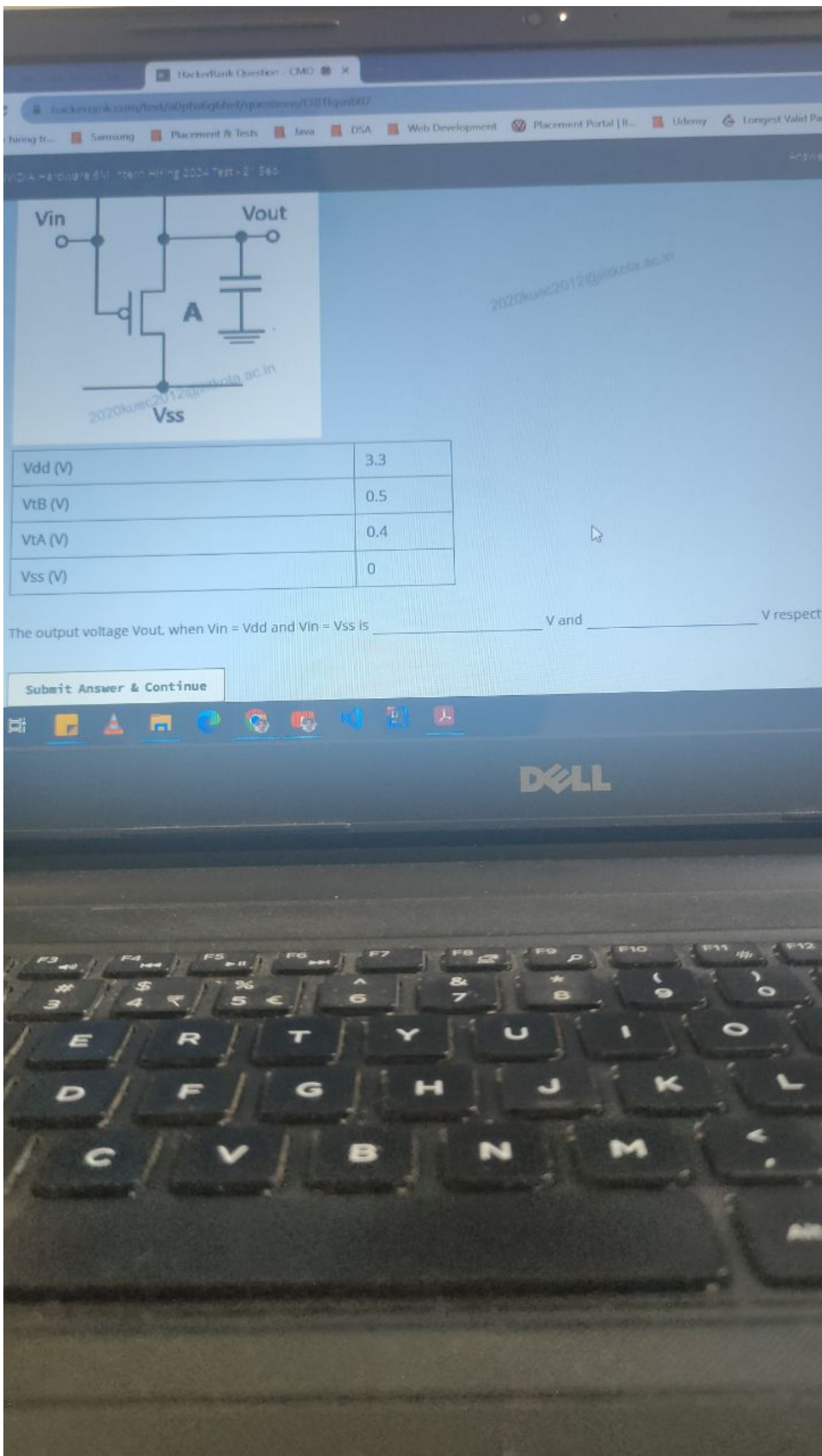
Note : if the answer is 8'd50 then just enter 50



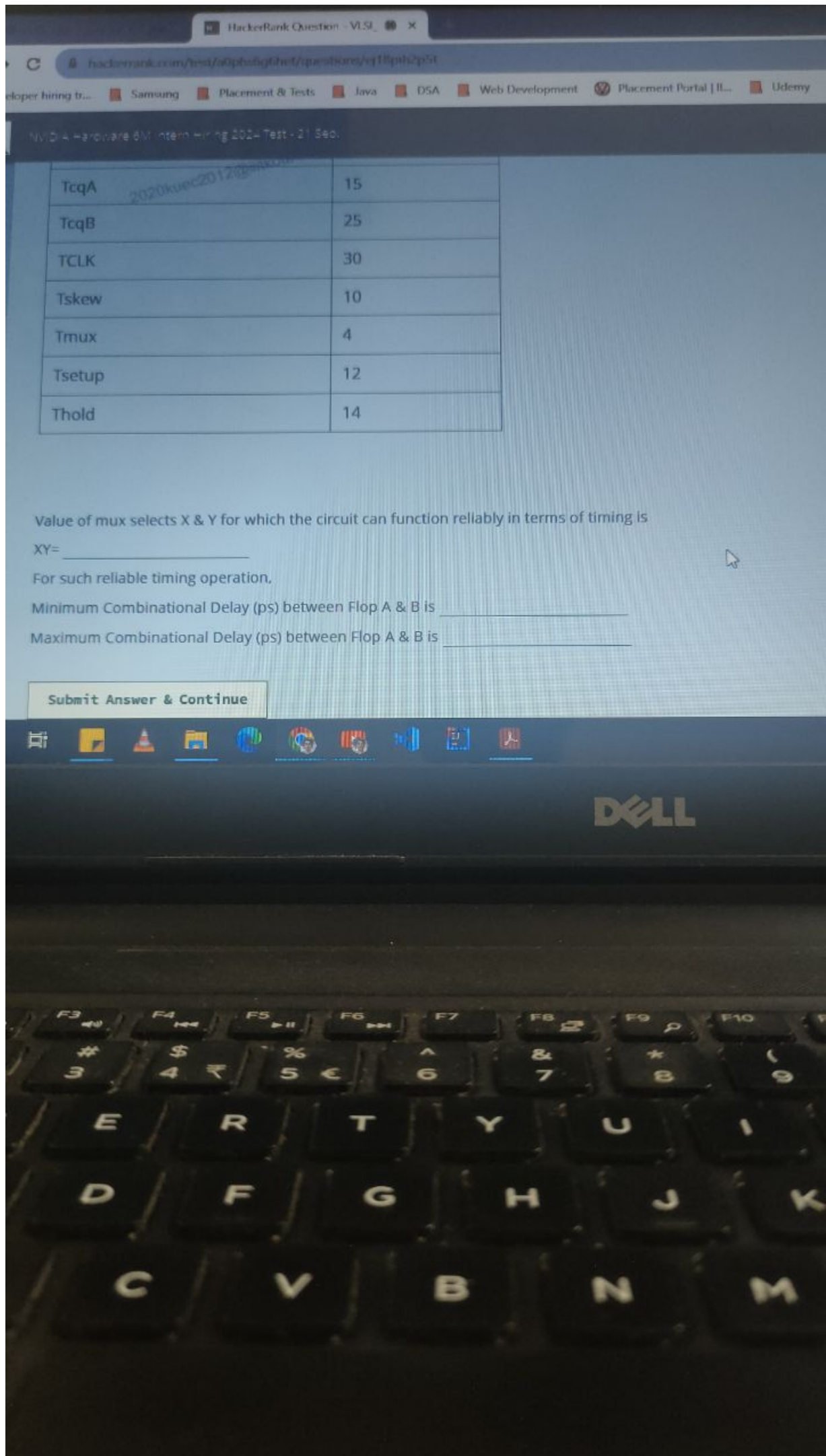












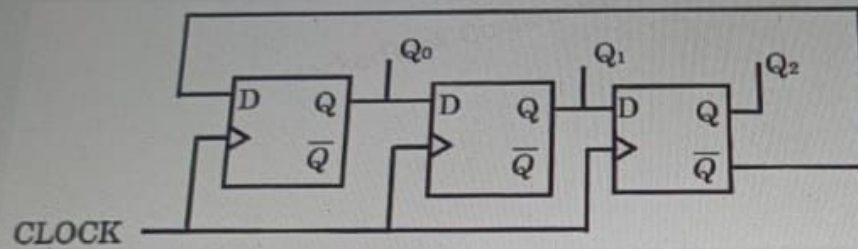


Complete the blanks in the following question with the appropriate answer.

ALL



Assume that the flops are reset initially, and value of (Q0, Q1, Q2) is (0,0,0) at the 0th clock pulse



|   |    |
|---|----|
| M | 23 |
| N | 45 |

What is the value of (Q0, Q1, Q2) at

(a) Mth clock pulse : \_\_\_\_\_ ;

(b) Nth clock pulse : \_\_\_\_\_ ;

Note: Enter the 3 bits without any coma, for example if the answer is 1,1,0; enter 110

Submit Answer & Continue

HackerRank Question - VLSI

hackerrank.com/test/afphs6q6het/questions/eg18pht2p6t

Developer hiring tr... Samsung Placement & Tests Java DSA Web Development Placement Portal | IL... Ude

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19. VLSI\_Timing\_Dynamic\_timing\_path

Complete the blanks in the following question with the appropriate answer.

Analyze the below circuit for timing parameters listed in table below:

The circuit diagram illustrates a dynamic timing path between two D flip-flops, A and B. Flip-flop A's output Q drives a chain of four buffers, each with a delay of  $T_{buf}$ . The output of this chain is connected to the D input of flip-flop B. Flip-flop B's output Q is connected back to the D input of flip-flop A. A clock signal CLK is applied to both flip-flops, with a skew  $T_{skew}$  between their clock inputs. The setup and hold times for both flip-flops are  $T_{setup}$  and  $T_{hold}$ . The propagation delays for flip-flops A and B are  $T_{cqA}$  and  $T_{cqB}$  respectively. A multiplexer with inputs 00, 01, 10, and 11 is shown, with outputs X and Y. The output of the buffer chain is connected to input 10 of the multiplexer.

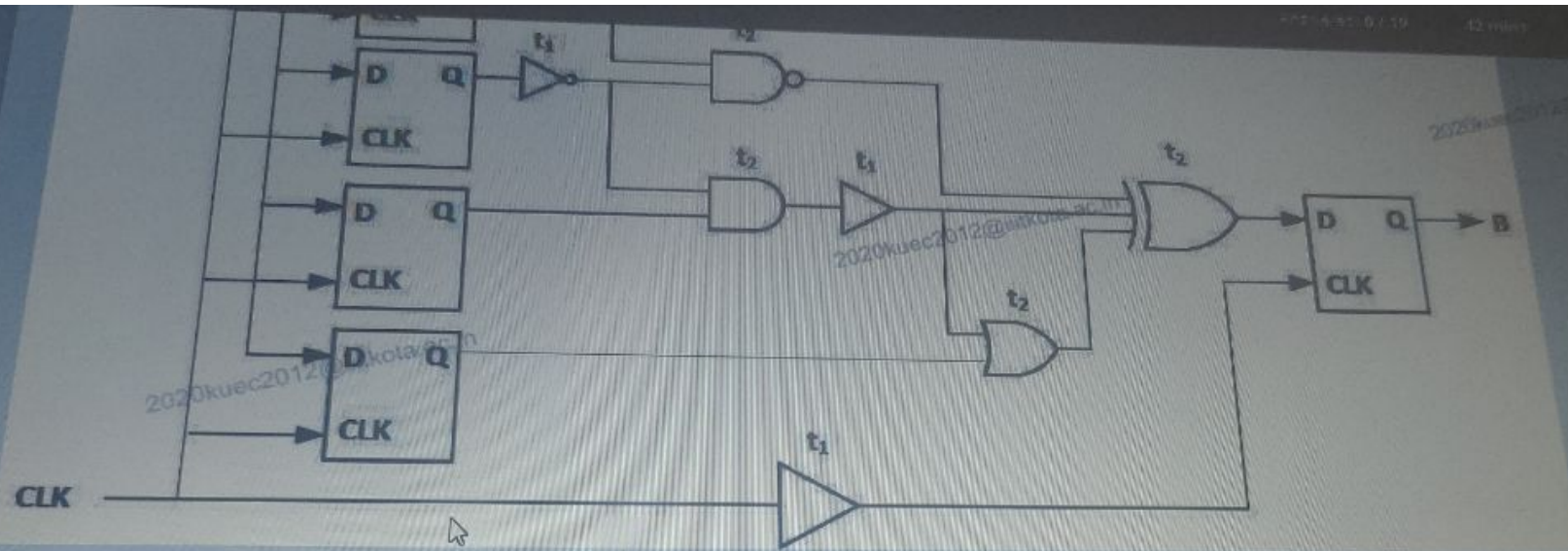
Note:  $T_{buf}$  represents the delay of each buffer.

| TIMING PARAMETER | VALUE (ps) |
|------------------|------------|
| $T_{buf}$        | 1          |

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V E R T Y U I O P  
D F G H J K  
C V B N M





|  | min<br>delay<br>(ns) | max<br>delay<br>(ns) |
|--|----------------------|----------------------|
|  |                      |                      |



Complete the blanks in the following question with the appropriate answer.

Design operating point changes from <frequency  $x_f$ , Voltage  $yV$ > to <frequency  $p_f$ , voltage  $qV$ >, where:

|   |    |
|---|----|
| x | 10 |
| y | 2  |
| p | 8  |
| q | 10 |

Dynamic power is multiplied by a factor of (enter number) : \_\_\_\_\_

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|          | min<br>delay<br>(ns) | max<br>delay<br>(ns) |
|----------|----------------------|----------------------|
| t1       | 2                    | 3                    |
| t2       | 3                    | 4                    |
| tck to q | 9                    | 11                   |
| tsetup   | 4                    |                      |
| thold    | 2                    |                      |

Minimum Clock Period the circuit can operate is \_\_\_\_\_ ns

Submit Answer & Continue

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Find the output of foo().

**Important:** Don't put spaces between commas and numbers in the result

```
void foo() {  
    int a[5] = {v, w, x, y, z};  
    int i, j, m, n;  
  
    i = ++a[1];  
    j = a[1]++;  
  
    m = a[i++];  
    n = ++a[j];  
  
    printf("%d,%d,%d", i, m, n);  
}
```

| v | w | x  | y  | z  |
|---|---|----|----|----|
| 5 | 2 | 15 | 20 | 25 |

Answer: \_\_\_\_\_

Submit Answer & Continue



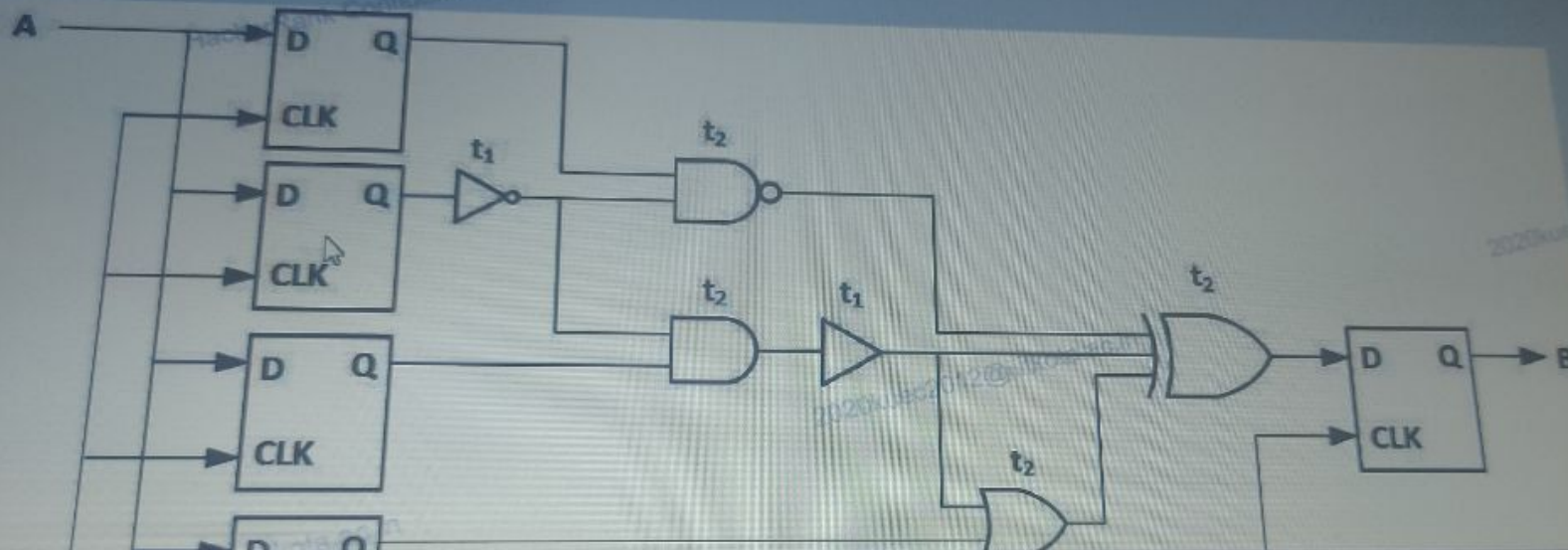
DELL



Complete the blanks in the following question with the appropriate answer.

Find the minimum clock period that is required for operating the following circuit. Please provide the answer in ns.

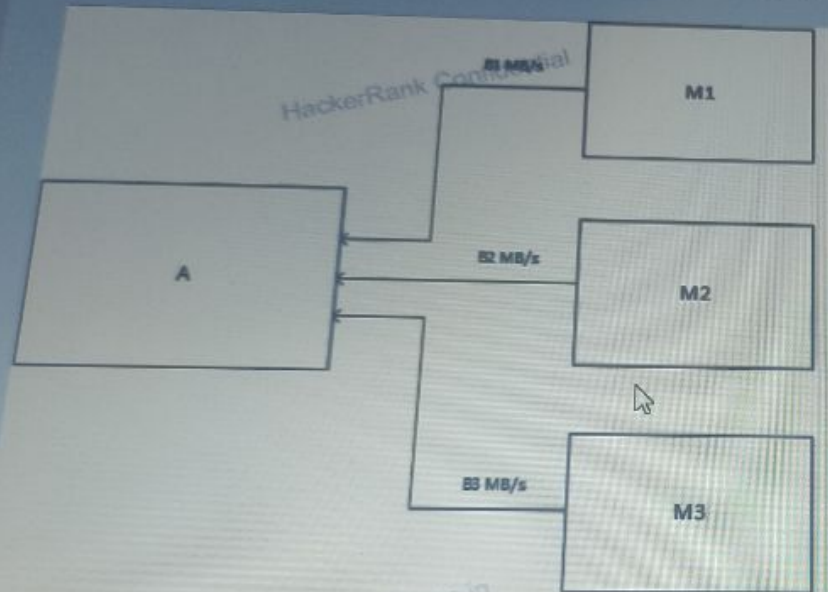
Note - Setup Time, Hold time & Clock-to-q delays are the same for all flip flops. All single input combinational gates have same delay of  $t_1$  and all multi-input combinational gates have same delay of  $t_2$ .



### 7. Mem-BW-calculation

Complete the blanks in the following question with the appropriate answer.

IP A reads data from 3 different memories as shown above, with bandwidth  $B_1, B_2$  and  $B_3$  MB/s respectively.



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## 9. Coding C Static type

Complete the blanks in the following question with the appropriate answer.

what is the value of j at the end of the execution of the following C program ?

```
#include <stdio.h>
```

```
int incr (int index)
```

```
{
```

```
static int count = 0;
```

```
count = count + index;
```

```
return (count);
```

```
}
```

```
main ()
```

```
{
```

```
int i,j;
```

```
int k; // Assume k is initialized to below value
```

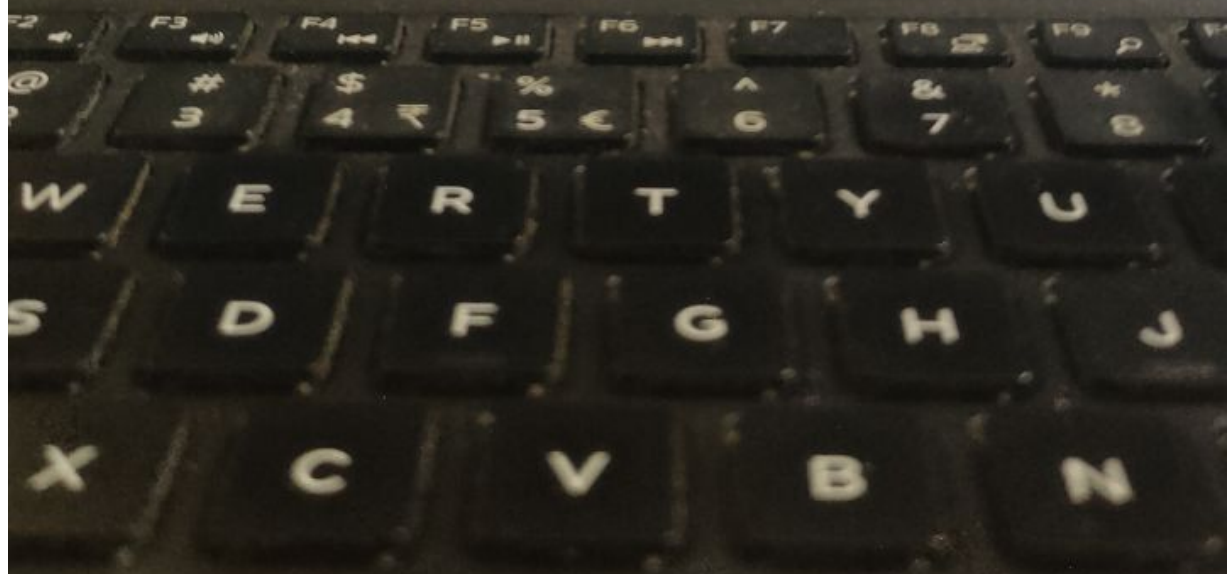
```
for (i = 0; i <= k; i++)
```

```
j = incr(i);
```

```
}
```



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```
int main()
{
    f(&i, &j);
    printf("i=%d j=%d\n", i, j);
    getchar();
    return 0;
}
```

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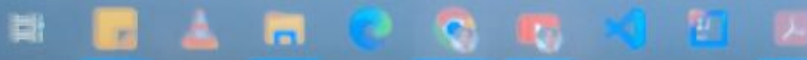
Values:

|   |   |
|---|---|
| i | 3 |
| j | 7 |
| k | 8 |

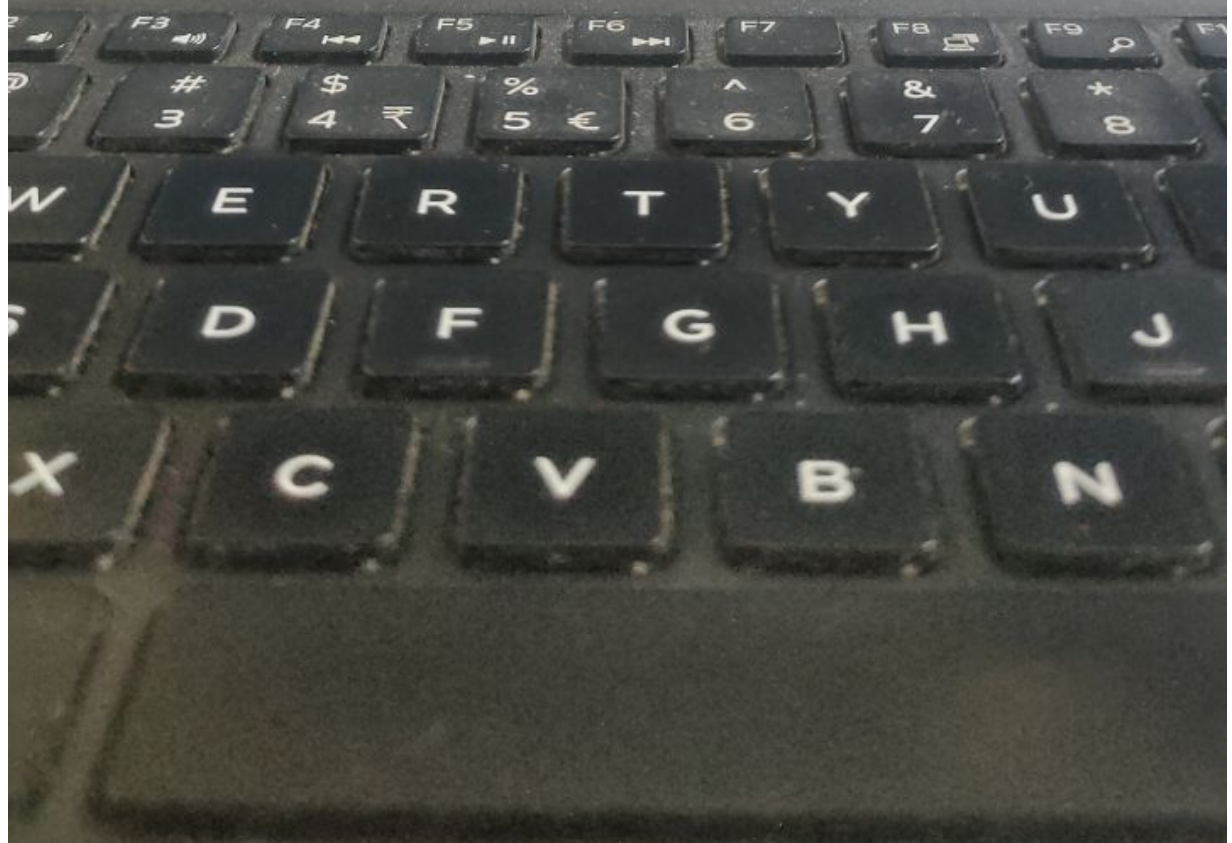
i = \_\_\_\_\_ :

j = \_\_\_\_\_ :

Submit Answer & Continue



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Complete the blanks in the following question with the appropriate answer.

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What is the output of the following program?

```
#include<stdio.h>
```

```
void f(int *p, int *q)
```

```
{
```

```
  p = q;
```

```
  *p = k;
```

```
}
```

```
int i, j, k; // assume that they are initialized to below values
```

```
int main()
```

```
{
```

```
  f(&i, &j);
```

```
  printf("i=%d j=%d \n", i, j);
```

```
  getchar();
```

```
  return 0;
```

```
}
```

Values:

|   |   |
|---|---|
| i | 3 |
|---|---|



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Given the below parameters:

| Parameters |    |
|------------|----|
| B1         | 30 |
| B2         | 90 |
| B3         | 80 |

Q1. The maximum possible bandwidth is \_\_\_\_\_ MB/s

Q2. The total data required by A is stored equally among the 3 Memories. The maximum data read Bandwidth possible for IP A is \_\_\_\_\_ MB/s

Q3. For achieving the BW in Q1, \_\_\_\_\_ % of total data should be in M1

Submit Answer & Continue

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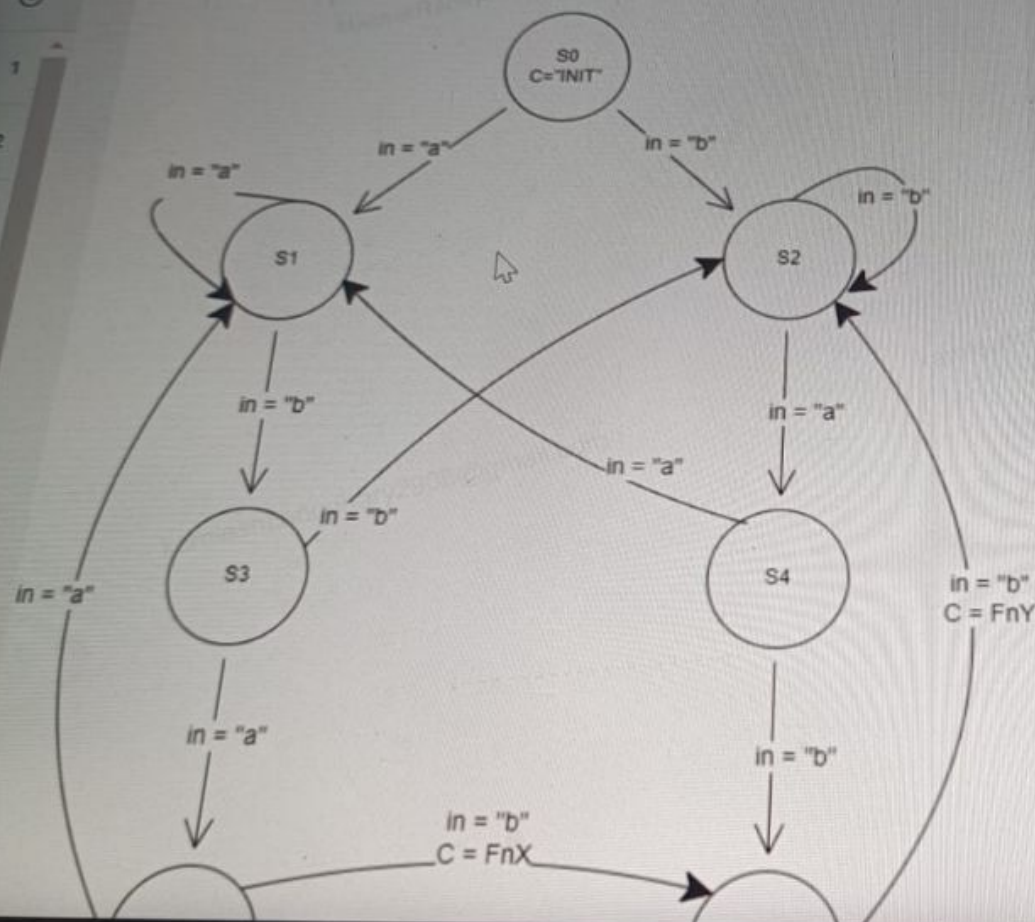


Complete the blanks in the following question with the appropriate answer.

Given below FSM which works on a single bit input "in", and generates output in the form of an 8 bit state variable "C".

The value of C is initialized to "INIT" in the idle state S0.

F<sub>n</sub>X and F<sub>n</sub>Y are operations performed on the variable C at the FSM transitions S5 -> S6 and S6 -> S2 respectively (as shown in the state diagram)

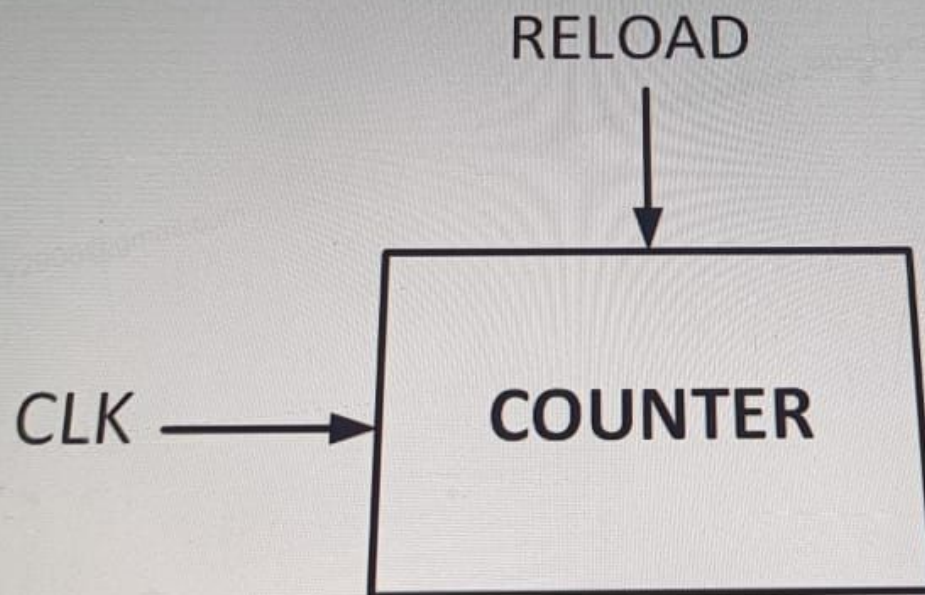




Complete the blanks in the following question with the appropriate answer.

The counter shown in the diagram below has the following characteristics:

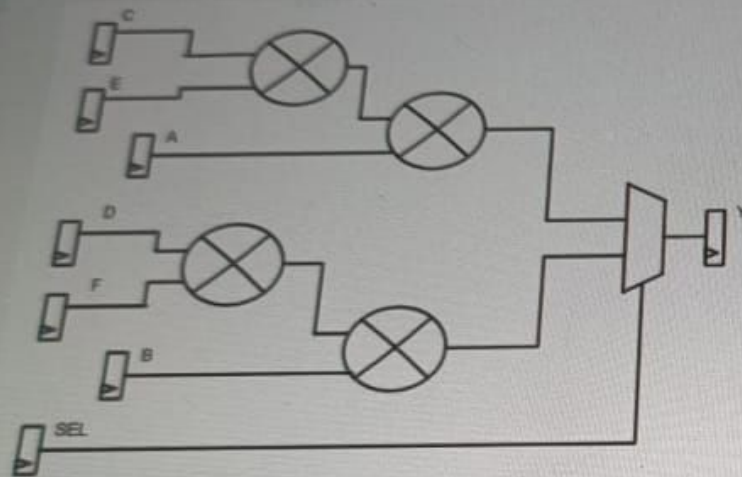
1. Counter increments by value "A" every "B" clock cycles.
2. When "RELOAD" signal in the diagram is asserted, the counter gets initialized to value "0"
3. The RELOAD signal gets asserted every "C" clock cycles and stays asserted for one clock.





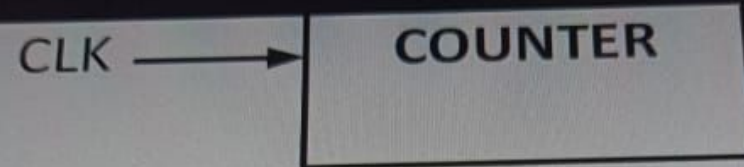
Complete the blanks in the following question with the appropriate answer.

The following block diagram shows an implementation of a computational logic.  
The area occupied by different components are shown below.  
Redesign the block diagram to achieve the same functionality with the minimum possible area.  
What is the total area of your design after minimization?



|                               |      |
|-------------------------------|------|
| Multiplier Area (sq. microns) | 2500 |
| Mux Area (sq. microns)        | 100  |
| Register Area (sq. microns)   | 100  |

Total area of the design after minimization (sq. microns) : \_\_\_\_\_

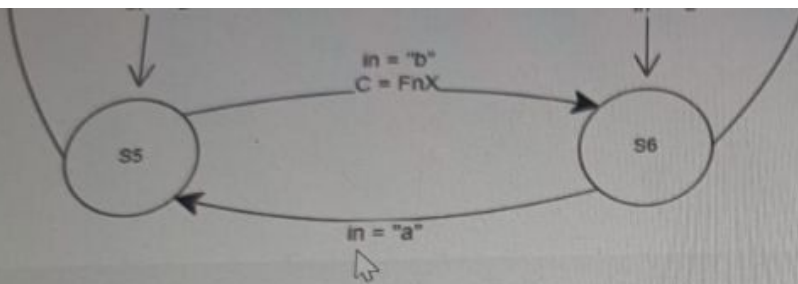


| Parameters |    |
|------------|----|
| A          | 4  |
| B          | 2  |
| C          | 8  |
| X          | 40 |
| Y          | 70 |

What is the counter value sampled at clock-cycle # X: \_\_\_\_\_

What is the counter value sampled at clock-cycle # Y: \_\_\_\_\_

Submit Answer & Continue



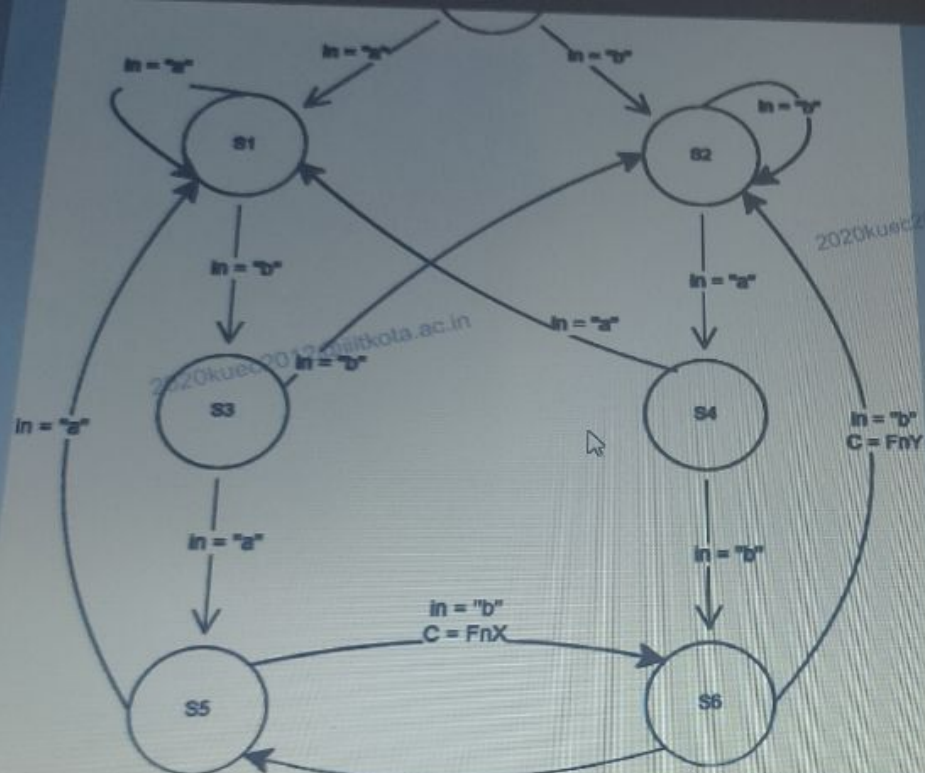
|      |                |
|------|----------------|
| a    | 0              |
| b    | 1              |
| FnX  | C++            |
| FnY  | C++            |
| INIT | 8'd50          |
| SEQ  | 10101010101010 |

For given parameters and SEQ, starting with leftmost bit identify the value of output C = \_\_\_\_\_

Note : if the answer is 8'd50 then just enter 50

Submit Answer & Continue





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## 2. Dynamic Power - Voltage Frequency

ALL



Complete the blanks in the following question with the appropriate answer.

Design operating point changes from <frequency  $x_f$ , Voltage  $yV$ > to <frequency  $p_f$ , voltage  $qV$ >, where:

|     |   |
|-----|---|
| $x$ | 2 |
| $y$ | 6 |
| $p$ | 4 |
| $q$ | 6 |

Dynamic power is multiplied by a factor of (enter number) : \_\_\_\_\_

Submit Answer & Continue

Complete the blanks in the following question with the appropriate answer.

Raya & Toya are running towards each other. The speed of the Raya is  $K1$  km/hr and the speed of Toya is  $K2$  km/hr. Raya throws a ball towards Toya when the distance between them is  $D$  meters. Once the ball reaches Toya, Toya immediately throws it back to Raya and so on. Each of them throws the ball at a speed of  $B$  km/hr.

|    |     |
|----|-----|
| K1 | 3   |
| K2 | 4   |
| D  | 130 |
| B  | 6   |

For the given values of  $K1$ ,  $K2$ ,  $D$ ,  $B$ , the total distance travelled by the ball till Raya and Toya cross each other is \_\_\_\_\_ meters.

[Note: Please roundoff answer to the nearest integer. For example :

if the answer is 3.5 m, enter 4

If the answer is 3.49 m, enter 3]

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## 12. Vowels never together

Complete the blanks in the following question with the appropriate answer.

Come up with the number of words that can be formed from the letters of word W (listed below) so that the vowels never come together.

W

TEGRA

Number of words \_\_\_\_\_

Submit Answer & Continue



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## 10. coding-c-array-incs

Complete the blanks in the following question with the appropriate answer.

Find the output of `foo()`.

**Important:** Don't put spaces between commas and numbers in the result

```
void foo() {  
    int a[5] = {v, w, x, y, z};  
    int i, j, m, n;  
  
    i = ++a[1];  
    j = a[1]++;  
  
    m = a[i++];  
    n = ++a[j];  
  
    printf("%d,%d,%d", i, m, n);  
}
```

| v | w | x  | y  | z  |
|---|---|----|----|----|
| 5 | 2 | 15 | 20 | 25 |

Answer:



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## 16. Processor Throughput

Complete the blanks in the following question with the appropriate answer.

A hypothetical CPU has at any point of time an average of  $N$  outstanding requests to memory. The Memory can transfer a max of  $X$  Bytes per mem cycle. The average time of each request is  $M$  core cycles, and each request is of  $A$  Bytes. There are 2 clock domains in the system. CPU domain:  $F_c$  Ghz. Memory Domain:  $F_m$  Ghz. Assume all buffers in system are sufficiently sized.

|                                         |     |
|-----------------------------------------|-----|
| Outstanding Requests $N$                | 70  |
| Memory Bytes per mem cycle $X$          | 128 |
| Request size $A$ Bytes                  | 64  |
| $F_c$ - CPU freq                        | 3   |
| $F_m$ - Memory Freq                     | 2   |
| Average time of each request $M$ cycles | 100 |

The Maximum Memory Bandwidth supported is \_\_\_\_\_ GBps

Note : Enter 1 decimal place and round it to nearest value. Ex: If the value is 2.63 Mbytes then type 2.6. If answer is 4, then just type 4 with no decimal.

